

MAT1002 Lecture 25,

Outline

- Surface integrals
 - ↳ Flux
- Curl of a 3D vector field

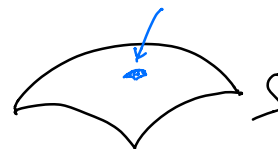
Surface Integrals

Let S be a smooth surface. Suppose that the planar density of S at (x, y, z) is given by $g(x, y, z)$, say kg/m^2 . Then the mass of S is given by the **surface integral** of g ,

$$\iint_S g(x, y, z) d\sigma,$$

where $\iint_S g(x, y, z) d\sigma$ is the limit of $\sum_k g(x_k^*, y_k^*, z_k^*) \Delta\sigma_k$ as the areas of the small regions ($\Delta\sigma_k$) goes to 0.

$$\text{Mass} \approx g(x_k^*, y_k^*, z_k^*) \Delta\sigma_k$$



If S is in **parametric form** given by $(x, y, z) = \mathbf{r}(u, v)$, then

$$\iint_S g(x, y, z) d\sigma = \iint_R g(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| du dv,$$

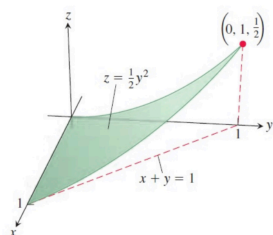
where R is the parameter domain.

If S is in **explicit form** $z = f(x, y)$, where f is defined on R , then

$$\iint_S g(x, y, z) d\sigma = \iint_R g(x, y, f(x, y)) \sqrt{f_x^2 + f_y^2 + 1} dx dy.$$

Example

Evaluate $\iint_S \sqrt{x(1+2z)} d\sigma$ on the portion of the surface $z = y^2/2$ over the triangular region $R: x \geq 0, y \geq 0, x + y \leq 1$ in the xy -plane.

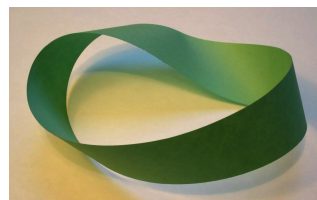
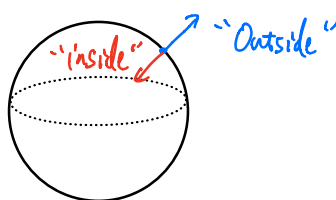
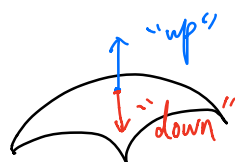


This is example 16.6.4.

Orientable Surfaces

Roughly speaking, these are surfaces with two sides.

e.g.



Not all surfaces are orientable. e.g., the Möbius strip.

In this course, we will only talk about orientable surfaces.

Flux: Motivation

Suppose that some fluid with constant density ρ passes through some planar region D with area A , at a constant velocity $\mathbf{v}$ perpendicular to D . The **mass flow rate** of the fluid across D is given by

$$\rho v A,$$

and it measures the mass of fluid passing through D per unit time.

If the fluid passes through D with a velocity $\mathbf{v}$ which is not perpendicular to D , then we need to consider the component of $\mathbf{v}$ in the direction of $\mathbf{n}$, where $\mathbf{n}$ is a unit normal vectors of D . In this case, the mass flow rate is equal to

$$\rho(\mathbf{v} \cdot \mathbf{n})A.$$

If $\mathbf{v}$ is a velocity field of some fluid, which has density function $\rho(x, y, z)$, then the **mass flow rate** of the fluid across an orientable surface S (which is a surface that has "two sides") is given by

$$\iint_S \rho \mathbf{v} \cdot \mathbf{n} \, d\sigma := \iint_S \rho(x, y, z) \mathbf{v}(x, y, z) \cdot \mathbf{n}(x, y, z) \, d\sigma.$$

Here $\vec{n}$ has a fixed orientation (i.e., always points to the same side of S).

A more general form of these integrals is given below.

Definition

Let $\mathbf{F}$ be a continuous vector field in $\mathbb{R}^3$, defined on an orientable surface S , with a unit normal vector $\mathbf{n}$ specifying the orientation of S . The **surface integral of $\mathbf{F}$ over S** , or the **flux of $\mathbf{F}$ across S** , is defined to be

$$\iint_S \mathbf{F} \cdot \mathbf{n} d\sigma.$$

- $\vec{r} = \vec{r}(x, y, z)$
- $\vec{r}$ specifies a side of the surface and the measuring direction of the flux.

e.g.



Flux: Computation

If S is a surface with parametrization $\mathbf{r}(u, v)$, then the unit normal vectors to S at a point are

$$\mathbf{n} = \pm \frac{\mathbf{r}_u \times \mathbf{r}_v}{|\mathbf{r}_u \times \mathbf{r}_v|}.$$

Here, the sign is determined by the context or question.

Suppose that the flux direction is given by the one with "+". Then

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} d\sigma &= \iint_R \mathbf{F} \cdot \frac{\mathbf{r}_u \times \mathbf{r}_v}{|\mathbf{r}_u \times \mathbf{r}_v|} |\mathbf{r}_u \times \mathbf{r}_v| du dv \\ &= \iint_R \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) du dv. \end{aligned} \quad \textcircled{1}$$

$\underbrace{\hspace{1.5cm}}_{d\vec{\sigma}}$

If S is given by $z = f(x, y)$, $(x, y) \in R$, then by using x and y themselves being parameters, then

$$\vec{r}_x = \langle 1, 0, f_x \rangle, \quad \vec{r}_y = \langle 0, 1, f_y \rangle \quad \text{and} \quad \vec{r}_x \times \vec{r}_y = \langle -f_x, -f_y, 1 \rangle,$$

and the formula ① above becomes

$$\begin{aligned} \iint_S \vec{F} \cdot \vec{n} \, ds &= \iint_R \vec{F} \cdot (\vec{r}_x \times \vec{r}_y) \, dx \, dy \\ &= \iint_R (-M f_x - N f_y + P) \, dx \, dy \end{aligned}$$

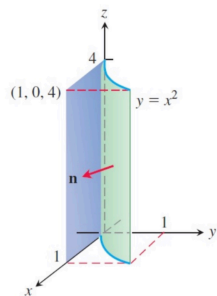
Example

Find the flux of $\mathbf{F} := \langle yz, x, -z^2 \rangle$ across the surface

$$S := \{(x, y, z) : y = x^2, 0 \leq x \leq 1, 0 \leq z \leq 4\},$$

Example 16.6.5

in the direction of $\mathbf{n}$ indicated in the figure.



e.g. Find the inward flux of $\vec{F} = \langle z, y, x \rangle$ across the sphere $x^2 + y^2 + z^2 = 1$.

Ans : $-4\pi/3$.

Curl

Definition

The **curl** of a vector field $\mathbf{F} = \langle M, N, P \rangle$ is defined to be

$$\text{curl } \mathbf{F} := \underbrace{\left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z} \right)}_{\text{circulation density projected on } yz\text{-plane}} \mathbf{i} + \underbrace{\left(\frac{\partial M}{\partial z} - \frac{\partial P}{\partial x} \right)}_{\text{" } xz\text{-plane}} \mathbf{j} + \underbrace{\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)}_{\text{" } xy\text{-plane}} \mathbf{k}.$$

Remark

If we think of ∇ as the symbolic vector $\langle \partial/\partial x, \partial/\partial y, \partial/\partial z \rangle$, then the curl can be remembered as the symbolic expression

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F}.$$

Example

Find the curl of $\mathbf{F} := \langle x^2 - z, xe^z, xy \rangle$.

Example 16.7.1